University of Minnesota
Driven to Discover

DRIVE-SAFE

Remote and Cooperative Autonomous Driving in Dynamic Environments using 5G/NextG Technology

MnRI Showcase
Student Poster Submission

1 Motivation

Edge cases beyond AV capabilities
construction, accidents, parking lots, rural roads

Need high bandwidth and low latency
for real-time teleoperation

Handover events can degrade teleoperation
performance due to mobility

2 System Architecture

Autonomous Vehicle (MnCAV – Chrysler Pacifica)



4 Cameras + LiDARs + IMU + GNSS

Vehicle sensor data
(e.g., camera feeds, LiDAR)



Uplink
(high bandwidth)

5G/NextG (O-RAN)

Command & control
(steering, throttle, brake)

Downlink
(low latency)

Remote Operator Station



3 Key 9-Month Demo Results

100%
fewer "ping-pong" and unnecessary handovers
Significantly improved tail latency and more stable connectivity during vehicle movement

~40%
lower tail latency
Reducing high-delay events due to mobility and handovers

~28%
lower average end-to-end video frame latency
Faster *uplink* sensor data delivery, improving responsiveness and situational awareness

Reduced jitter
and improved reliability
Smoother *uplink* video streaming and more stable *downlink* vehicle-control communication

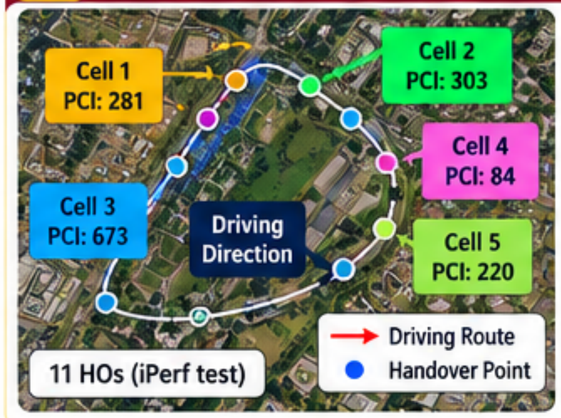
4 Driving Scenarios (POWDER Testbed)



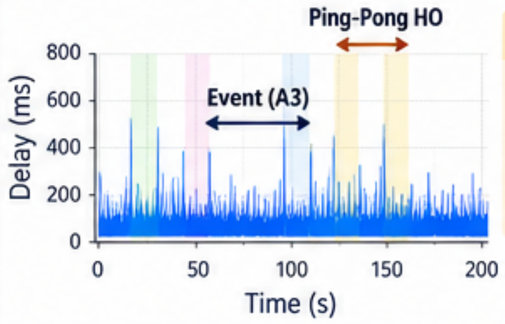




5 Handover Analysis (Field Test)



Handover Impact on Per-Frame Delay



11 HOs observed
- PRACH warning
- RAR timeouts
- A3 = 6 dB
- TTT = 100 ms
- Zero hysteresis

6 5GAA Remote Driving Latency Requirements

Autonomous Vehicle (w/ cameras + LiDARs)



UL: 40–60 ms
Sensor data delivery

5G/NextG (O-RAN)

DL: 15–20 ms
C&C data delivery

Teleoperation Station



End-to-end uplink (camera to display)	End-to-end downlink (C&C data)
< 100 ms	< 20 ms
Uplink network-level latency	Downlink network-level latency
< 60 ms	< 15 ms

Source: 5GAA, "Tele-operated driving (ToD): System requirements analysis and architecture," 2021.

7 Key Takeaways

Mobility-aware link adaptation
improves network latency, stability and reliability

Trajectory-aware handover avoidance
reduces high-delay events and video interruptions

Enables safe remote driving
in dynamic and real-world environments